

AMD's on-device AI: Just how good is it?

Let's explore what the Ryzen AI 300 series can do for real-world users

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With the Ryzen AI series, AMD has brought out the second generation of mobile processors with inbuilt NPUs, built on some of the IPs acquired via Xilinx. Since raw hardware capability means little without tangible use cases, let's explore what the Ryzen AI 300 series can actually do for real-world users. We look at applications like local LLMs running via LM Studio, and voice-to-text models—all without hitting the cloud. Using the Lenovo ThinkPad T14 Gen 6 as our test platform, we evaluate how AMD's new AI engine stacks up against the competition, particularly Intel's Core Ultra chips.

Test Bench

To evaluate the real-world capabilities of the Ryzen AI 7 PRO 360, we used a thoughtfully configured test platform: the Lenovo ThinkPad T14 Gen 6. This business-class laptop strikes a balance between portability, durability, and performance—making it an ideal candidate for gauging how AMD's on-device AI performs in professional workflows.

At the heart of this machine is the AMD Ryzen AI 7 PRO 360, featuring eight Zen 4 CPU cores, integrated AMD Radeon 880M graphics, and a second-generation XDNA 2 NPU. This

particular model includes 64 GB of LPDDR5X memory soldered directly to the motherboard and rated for 7500 MT/s, which not only aids high-throughput AI tasks but also ensures excellent responsiveness across multitasking scenarios.

Storage is handled by a fast 512 GB PCIe SSD, while connectivity includes a Qualcomm NCM825 module offering Wi-Fi 7 (2x2 BE) and Bluetooth 5.4. This ensures low-latency wireless networking—a relevant factor when evaluating hybrid AI use cases that may rely on cloud fallback or real-time collaboration tools.

Thermal design is another key aspect worth noting. The T14 Gen 6 stays true to ThinkPad traditions, maintaining quiet operation under sustained loads. During AI testing, fan noise remained controlled, and surface temperatures around the palm rest stayed comfort-

able. This matters when running local LLMs or Whisper-based transcription in the background—tasks that can engage the CPU, GPU, and NPU for long.

In short, the ThinkPad T14 Gen 6 offers a robust and balanced platform for evaluating on-device AI. It's neither an ultrabook with limited thermals nor a bulky workstation—just a well-rounded, enterprise-class machine capable of demonstrating the full potential of the Ryzen AI 7 PRO 360 in both consumer and professional contexts.

On-Device AI

The true measure of any AI-enabled processor lies in what it enables users to actually do. With the Ryzen AI 300 series, AMD has focused on offloading specific machine learning tasks from the CPU and GPU to the integrated NPU, helping reduce latency and power consumption while freeing up system resources for other workloads. In our testing of the Ryzen AI 7 PRO 360 on the Lenovo ThinkPad T14 Gen 6, we explored a couple of categories of practical on-device AI applications: real-time video enhancement, local large language models (LLMs), and speech-to-text transcription. Each showcases a different strength of the NPU and broader platform.

LM Studio with Local LLMs

We also tested the Ryzen AI 7 PRO 360 using LM Studio, a GUI-based interface for running local large language models such as LLaMA 3 and Mistral. The benefit of using LM Studio is that it abstracts away much of the complexity involved in deploying transformer models locally and provides direct GPU/NPU acceleration where supported.

Running a 7B model (quantised to 4-bit) entirely on-device, the Ryzen AI chip was able to deliver inference rates of up to 7.42 tokens per second, with prompt processing speeds of 26.33

